

Classification of Patients with Breast Cancer using Neighbourhood Component Analysis and Supervised Machine Learning Techniques

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Abstract—Breast cancer is considered one of the leading causes of death among women. In Morocco, the ministry of health reports over 40,000 new cases each year. When lifestyle can be a preventive pattern, early detection remains a factor of a huge impact on the mortality of the diseases. Machine learning (ML) algorithms offer an alternative to breast cancer standard techniques of prediction, or at least can assist radiologists in their reasoning flow and thus saving many females and some males from breast cancer biopsy. The present study represents a benchmarking of different ML models. The research applies and compares four machine learning algorithms (kNN, decision tree, Binary SVM, and Adaboost) to predict whether a patient has a malignant or a benign tumor. The machine learning techniques have been trained then tested on the Breast Cancer Wisconsin dataset. The datasets features are fed into feature selection model with Neighbourhood Components Analysis (NCA) to reduce the number of features and therefore decrease the complexity of the model. The predictive accuracy reached a 99.12% for the kNN model, the best predictive specificity obtained was 98.86% for the Binary SVM model and the highest predictive sensitivity obtained was up to one for both kNN and Adaboost models.

Keywords—Breast cancer, classification, features selection, machine learning, Neighbourhood Component Analysis.

I. INTRODUCTION

Breast cancer is one of the most frequently diagnosed cancers and the leading cause of women's death around the globe [1]. In 2018, over 2 million new cases of cancer diagnosed were estimated. Besides, breast cancer mortality remains higher in Eastern Europe and Africa [2], [3]. To prevent the spread of breast cancer, the Worldwide Health Organization (WHO) calls for the adoption of a healthy lifestyle. Early detection remains an essential component to reduce the mortality rate of breast cancer [4]. At its different stages, breast cancer diagnoses can use various imaging techniques such as mammography, positron emission tomography (PET), magnetic resonance imaging (MRI), Computed tomography (CT), and single-photon emission computed tomography (SPECT) [5].

Early detection has set a necessity for the use of machine learning techniques. Computer-Aided Diagnosis (CAD) systems have become part of routine clinical work worldwide used for breast cancer detection on mammograms at many screening sites and hospitals [6], [7]. Since 1980, CAD systems thrived through the years to become more independent and more accurate in terms of increasing true positive rate detection and decreasing incorrectly diagnosed cases [8], [9]. The classification of cancer-based CAD systems

may use as input statistics attributes from a data set which regroups different kinds of features such as dynamic features, textural features, and morphological features [10].

Machine learning (ML) techniques, such as K-Nearest Neighbors, and Decision Tree, are widely used to predict and pre-diagnose diseases like cancers, heart diseases [11], and hepatitis [12].

In this paper, four of the ML techniques, K-Nearest Neighbors, Decision Tree, Binary SVM, and AdaBoost, are adapted, then applied on a database of breast cancer Wisconsin dataset after features selection with Neighbourhood Component Analysis (NCA), to predict the severity of breast cancer [13]. This paper aims to provide a benchmarking on the four models and gives the best systems based on many performance evaluations measures such as accuracy, sensitivity, specificity, and others.

The present paper is arranged as follows: Section II presents an overview of some relevant related work in the field of breast cancer disease classification. Section III gives a global view of the dataset utilized and the ML algorithms used in this study. Section IV offers details of the implementation and the results obtained. We conclude this paper in section V.

II. RELATED WORK

Over the last few years, Breast cancer detection using ML-based systems has been the focus of many studies, many methodologies have already proven their ability to predict and classify breast cancer. However, this does not in the least restrict research in the field. On the contrary, scientists invest time and effort to develop various ML models tested on different datasets that seeks an effective prediction of cancer with higher accuracy [14] [15].

This section presents some of the research work-related to the breast cancer prediction among the type of data exploited, the predictive algorithms used, and the evaluation performance measurement.

Puja Gupta et al. [16] have compared six classifier algorithms (kNN, L_G, DT, RF, SVM and DL_ANN) used their hyperparameters on Wisconsin Breast Cancer Dataset (WBCD). Their results demonstrate that among the six algorithms, deep learning using Adam Gradient Descent Learning has shown a 98.24% accuracy in classifying the recurrence of the breast cancer disease.

Vikas Chaurasia et al. [17] developed a prediction model tested on Wisconsin Breast cancer dataset (BCWD) comparing three data mining algorithms (Naïve Bayes, RBF

Network, J48) and used 10-fold cross-validation methods to measure the performance of the three prediction models. Their results indicated that the Naïve Bayes is the best predictor with 97.36% accuracy.

Mohammed H. Tafish et al. [18], based on data collected from the Gaza strip Breast Cancer Data in Gaza Strip (BCDG), proposed a model that aims to help determine the degree of risk for the breast cancer disease and to allow practice for medical students. The model applies classification techniques such as SVM, ANN, and KNN on the collected breast cancer data, to predict the severity of breast cancer. After evaluation and testing using the mentioned classification techniques on breast cancer datasets, they obtained an accuracy of 77%.

III. MATERIALS AND METHODS

A. Dataset

In this study, we exploited the Breast cancer Wisconsin Dataset (BCWD). The dataset is multivariate, and it regroups a total of 569 instances. Its main task is to classify the tumor as benign or malignant; it contains 32 attributes or features. For 357 instances are; benign and 212 are malignant masses computed from a digitized image of a fine needle aspirate (FNA) of breast mass created at the University of Wisconsin [19]. From the 32 features, 2 attributes present information about the instances; the ID number and the diagnosis (malignant or benign). The other 30 features resulted from calculating 3 values (mean value, standard error and extreme value) of ten real-valued features (radius, Texture, perimeter, area, smoothness, compactness, concavity, concave points, symmetry, fractal dimension) that were computed for each image.

B. Neighborhood Component Analysis (NCA)

Feature subset is a mechanism that identifies irrelevant or redundant features to remove as many attributes as possible and thus selecting the most relevant ones. The literature presents a large amount of feature selection methods [20]. Neighbors Component analysis (NCA) is one of them. NCA is a statistical method used to maximize the accuracy of classification or regression algorithms, reduce the dimensions of the dataset, and offer better assimilation of the data in the ML context. The purpose of NCA is to ensure an optimal objective function (Eq.3) using a gradient-based optimizer. Unlike K-Nearest Neighbor, NCA selects randomly a neighbor and search for the expected vote for each class where each point i selects a neighbor point j with a probability of P_{ij} (Eq.1) with d_{ij} the distance between points i and j [21].

In a multiclass classification, let $S = \{(x_i, y_i), i = 1, 2, \dots, n\}$ be a labelled training data set, where $x_i \in R^p$ are feature vector, $y_i \in \{1, 2, \dots, c\}$ are the class labels with c the number of classes and n the number of observations.

The distance between the feature vector is calculated using the sum of their absolute difference multiplied by the weight of the vector. with d_{ij} (Eq.1) the distance between points x_i and x_j [21].

$$d_{ij} = \sum_{l=1}^p w_l^2 |x_{il} - x_{jl}| \quad (1)$$

where, w_l is the weight for the l^{th} feature.

Unlike K-Nearest Neighbor, NCA selects randomly a neighbor and search for the expected vote for each class where

each point x_i selects a neighbor point x_j with a probability of P_{ij} (Eq.2).

$$P_{ij} = \frac{k(d_{ij})}{\sum_{k \in C_i} k(d_{ik})}; P_{ii} = 0 \quad (2)$$

Where, $k(z) = \exp(-z/\sigma)$ is a kernel function and σ is the kernel width which influences the probability of each point being selected as the reference point.

The probability P_i represents the expected fraction of time that point i will be labelled in the right class denotes the set of points in the same class as i .

$$P_i = \sum_{j \in C_i} P_{ij} \quad (3)$$

The following expression is used to maximize the objective function.

$$f(w) = \frac{1}{n} \sum_{i=1}^n P_i \quad (4)$$

C. Proposed predictive system

The process to build a breast cancer classification-based system starts with finding data, as already mentioned, we used the BCWD. The dataset was split into training and testing data with a percentage of 80% and 20% respectively. The target is to classify the tumor as malignant or benign. After features selection with NCA, Four ML algorithms (AdaBoost, BSVM, DT and k-NN) were trained and then tested. To evaluate the models, we used the confusion matrix and other related metrics such as accuracy. The Receiver operating curve was another evaluation factor taken into consideration. Figure 1 presents the main stages of our proposed predictive system.

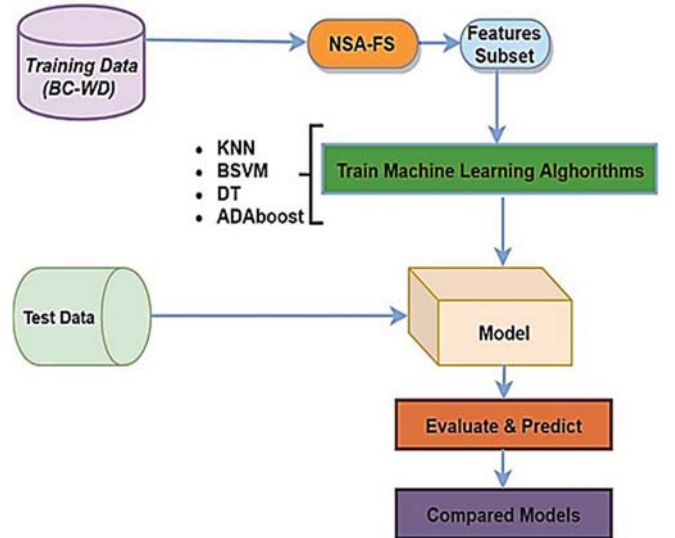


Fig. 1. Flowchart of the predictive model.

D. Performance evaluation methods

1) Confusion matrix for binary prediction

In supervised ML and specifically in statistical classification, a confusion matrix or an error matrix is a table used to assess the quality of a classification of an algorithm. The principle consists in comparing the instances in a predicted class (each row) with the instances in the reference class (each column) [22]. Table I present a confusion matrix with two rows and two columns that reports the number of false positives FP, false negatives FN, true positives TP, and true negatives TN, with:

- True positive (TP): This refers to really ill patients predicted as sick.
- False positive (FP): This refers to healthy patients predicted as sick.
- True negative (TN): This refers to healthy patients predicted as healthy.
- False negative (FN): This refers to sick patients predicted as healthy.

TABLE I. CONFUSION MATRIX FOR BINARY PREDICTION

Predicted output/labels (Ground truth)	Algorithm outputs/labels	
	Patient without disease (Negative/0)	Patient with disease (Positive/1)
Patient without disease (False/0)	TN	FP
Patient with disease (True/1)	FN	TP

2) Performance evaluation measures

To evaluate the performance of the algorithms and to concretize the achievement of the models, this study used six basic metrics (sensitivity, specificity, accuracy, Jaccard, Dice, and MCC) derived from the outcome of the confusion matrix. The sensitivity gauges the ability of the algorithm to correctly detect ill patients who do have the condition. The specificity measures the algorithm's ability to correctly identify healthy patients without a condition [23]. The accuracy measures the proportion of true results positives and negatives among the total number of cases examined. The MCC stands for Matthews's correlation coefficient, and it measures the quality of classification based on two classes. The Jaccard Similarity equals the size of the intersection on the union of the two classes, it is used to measure the similarity between them [24]. The Dice coefficient, just like Jaccard but in a more semi metrical way, is a statistic used to measure the similarity between two classes [25].

$$\text{Sensitivity} = \frac{TP}{FP+TP} \quad (5)$$

$$\text{Specificity} = \frac{TN}{FP+TN} \quad (6)$$

$$\text{Accuracy} = \frac{TP+TN}{TP+FP+FN+TN} \quad (7)$$

$$\text{MCC} = \frac{TP \cdot TN - FP \cdot FN}{\sqrt{(TP+FP) \cdot (TP+FN) \cdot (TN+FP) \cdot (TN+FN)}} \quad (8)$$

$$\text{Jaccard} = \frac{TP}{TP+FP+FN} \quad (9)$$

$$\text{Dice} = \frac{2 \cdot TP}{2 \cdot TP + FP + FN} \quad (10)$$

IV. RESULTS AND DISCUSSION

A. NCA for features selection

To detect relevant features from the original 32 attributes of the Wisconsin dataset, we used Neighborhood Component Analysis (NCA). We used the minimum lambda of 0.002313 that correspond to the minimum loss as it is represented on Figure 2.

The dimensions of our dataset have gone from 32 features to only 12 relevant attributes. Indeed as it is shown in Fig. 3, the features 1, 2, 4, 5, 6, 7, 8, 10, 11, 13, 16, 17, 18, 19, 20, 21,

27, 28, 30 have a weight that equals 0, which allows us to reduce the dimensions of our data set and remove all those irrelevant features.

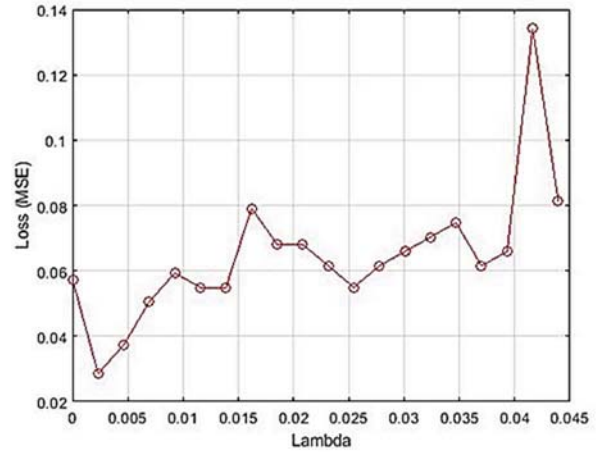


Fig. 2. Minimum loss in accordance to Lambda values.

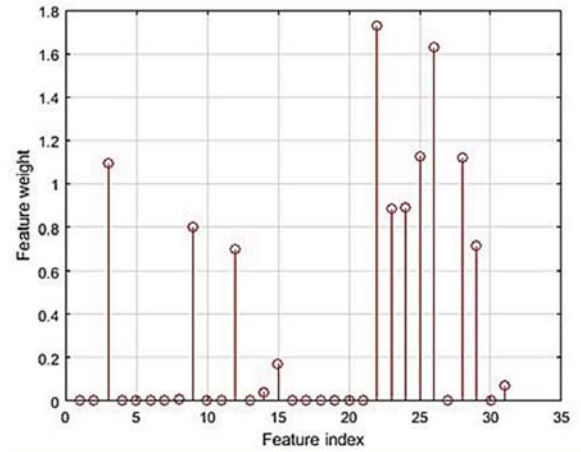


Fig. 3. Features weight in accordance to features index.

B. Training and algorithms best configuration

After preparing the dataset and dividing it into 80% for the training process and 20% for testing the models, we try to find the best parameters or the algorithm's configuration with the parameters that give the best metrics. For the k-NN, we varied k in a range of 1 to 40. The optimal model is obtained with $k=10$ by considering and calculating the Euclidian distance. For the decision tree model [26], the cross-validation loss was decreased using an optimize hyperparameters we repeated the same process for Binary SVM and for AdaBoost, we applied 200 cycles to reach the minimum error. Table II summarizes the best configuration of the KNN model.

TABLE II. PROPOSED K-NN ARCHITECTURE AND TRAINING PARAMETERS.

Architecture parameters	Value
K_{max}	40
k	10
Learn rate	1
Cross validation Loss kloss	0.0330
Resubstituting Loss rLoss	0.0308
Evaluation of the generated model quality E	0.8866

Table III present the best configuration parameters of DT, BinarySVM (BSVM) and ADABOOST models.

TABLE III. PROPOSED ARCHITECTURE AND TRAINING PARAMETERS FOR DT BSVM AND ADABOOST MODELS.

	<i>DT</i>	<i>BSVM</i>	<i>ADABOOST</i>
Best estimated feasible point	MinLeafSize: 28	BoxConstraint: 1.0769 KernelScale: 995.43	MinLeafSize: 21
Estimated objective function value	0.079206	0.32411	0.027505
Estimated function evaluation time (s)	0.0804	0.334028	23.5771

C. Comparison of testing results

Table IV shows, that after the features selection, among the four algorithms applied on the 114 patients of the testing data, k-NN gives the highest rate of TN indicating 88 healthy people correctly identified as healthy, 0 FP which mean not a single healthy people incorrectly identified as sick and only 1 FN which means 1 ill patient incorrectly identified as healthy. On the other hand, BSVM comes near the k-NN in terms of TN since it detected correctly 87 patients as healthy and 1 FN.

TABLE IV. CONFUSION MATRIX OF BREAST CANCER DISEASES PREDICTION USING ML ALGORITHMS TESTED ON 114 PATIENTS FROM BREAST CANCER WISCONSIN DATASET AFTER FEATURES SELECTION.

	<i>TN</i>	<i>FN</i>	<i>FP</i>	<i>TP</i>
AdaBoost	81	7	0	26
k-NN	88	1	0	25
DT	78	10	2	24
BSVM	87	1	5	21

TABLE V presents the values of the different evaluation metrics parameters such as the Accuracy, the Sensitivity, the Specificity, etc. obtained from the results of the matrix confusion of the four models.

These results show that k-NN is the best model in term of both accuracy and sensitivity, indeed the models accuracy is 99.12% and its sensitivity reach up to 1 which make the k-NN the most accurate algorithm from the four to correctly identify sick people, the best one to provide the proportion of true results among the number of testing patients, and to differentiate between healthy and sick patients.

In the other hand BSVM is the best model with the highest probability to correctly reject healthy patients since it gives the highest specificity of 98.86% among the four algorithms.

TABLE V. EVALUATION PERFORMANCE MEASURE OF ML ALGORITHMS TESTED ON 114 PATIENTS FROM BREAST CANCER WISCONSIN DATASET AFTER FEATURES SELECTION.

<i>Model</i>	<i>AdaBoost</i>	<i>KNN</i>	<i>DT</i>	<i>BSVM</i>
<i>Metrics</i>				
<i>Accuracy</i>	93.86	99.12	89.47	94.74
<i>Sensitivity</i>	1	1	92.31	98.86
<i>Specificity</i>	92.05	98.88	88.64	80.77
<i>MCC</i>	85.16	97.51	74.24	84.67
<i>Jaccard</i>	78.79	96.15	66.67	77.78
<i>Dice</i>	88.14	98.04	80.00	87.50

D. Results in terms of Receiver Operating Curve (ROC)

The Receiver Operating Curve or the ROC is a graph that presents a simpler alternative for evaluating a model then the confusion matrix. The y-axis displays the true positive rate (Sensitivity) while the x-axis shows the false positive rate (Specificity).

The AUC stands for the area under the curve and it allows to compare between different ROCs in a tangible way. AUC value of a classifier falls normally between the two values 0.5 and 1.

An AUC under 0.50 indicates a set of random data, which could not differentiate between true positive and false positive [27]. The higher the AUC is the better, in fact, an optimal classifier has a value of AUC close to 1.

Figures 4-7 depicts the ROC of each model while Table VI regroups the AUC values evaluating the corresponding ROC of the different models. It shows that the k-NN model have the highest AUC with a value of 0.9558.

TABLE VI. AUC FOR KNN, BSVM, DT AND ADABOOST MODELS.

<i>Model</i>	<i>kNN</i>	<i>BSVM</i>	<i>DT</i>	<i>AdaBoost</i>
<i>AUC</i>	0.9558	0.9140	0.9047	0.9366

• *k-NN*

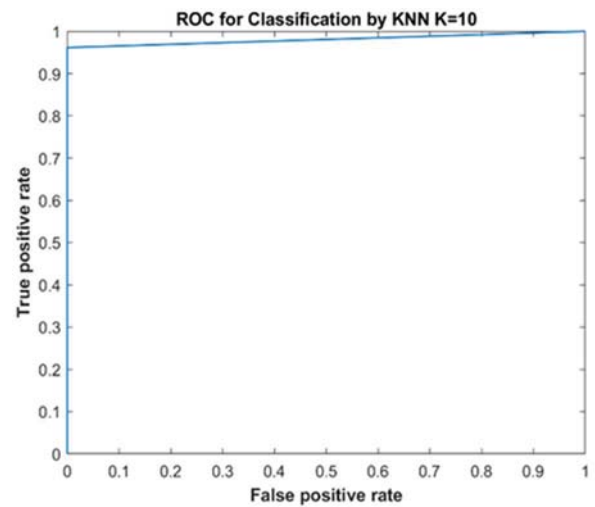


Fig. 4. ROC for classification via the kNN model.

• *BinarySVM*

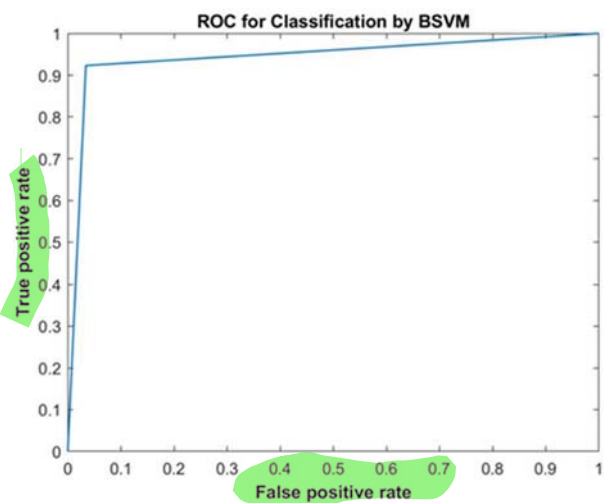


Fig. 5. Roc for classification by BSVM.

- *Decision Tree*

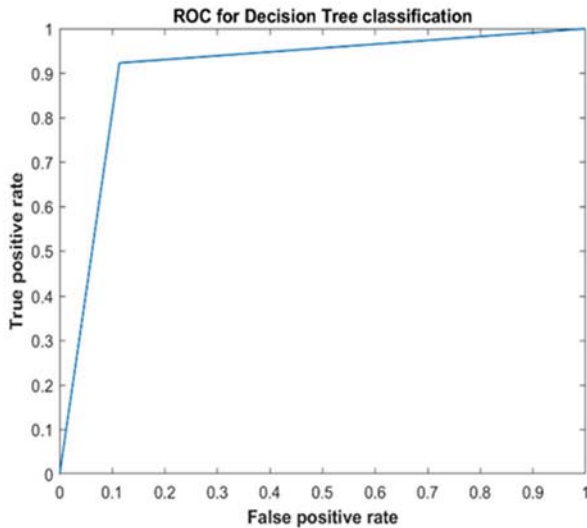


Fig. 6. Roc for classification by DT model.

- *ADABOOST*

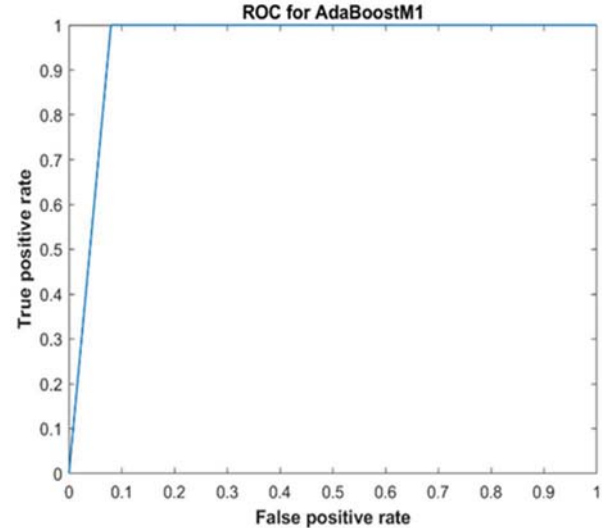


Fig.7. ROC for classification by ADABOOST model.

E. Discussion

This paper has examined 4 supervised machine learning techniques to present a benchmarking of the different models in breast cancer classifications. Based on the results of this study, k-NN has proven its power in the classification of breast cancer severity with NCA for feature selection. Still, the model does not lack reservations. Indeed k-NN is an instance-based algorithm, on the contrary to eager learning algorithms such as DT, it takes an important amount of computational time for classification and it depends on cross-validation or other techniques to choose k. As already mentioned in section II, Puja Gupta et al. [14] and Vikas Chaurasia et al. [15] used

the WBCD to train and test the different machine learning algorithms. They respectively registered an accuracy of 98% (DL-ANN) and 97% (for Naïve Bayes). Our results, on the other hand, show that k-NN is more precise in classifying breast cancer with an accuracy of over 99%. These results prove that features selection with NCA can have a tangible impact on the accuracy of the models for predicting breast cancer severity in general; more specifically on the k-NN model since based on our study, k-NN model is 0.04 more accurate than the k-NN model in [14]. In Table VII, we summarize the main results in Section II in the goal to give a comparison with our results.

TABLE VII. EVALUATION METRICS RESULTS.

Authors/Reference	Predictive Models	Used database	Performance metric		
			Accuracy	Sensitivity	Specificity
Puja Gupta et al. [13]	KNN	WPBC dataset	0.95	N/A	0.93
	L_G		0.95	N/A	0.95
	DT		0.95	N/A	0.95
	RF		0.97	N/A	0.97
	SVM		0.97	N/A	0.97
	DL ANN		0.98	N/A	0.98
Vikas Chaurasia et al. [14]	Naïve Bayes	Breast Cancer Wisconsin dataset	0.97	0.97	0.97
	RBF Networks		0.96	0.97	0.96
	J48		0.93	0.93	0.90
M. H. Tafish et al. [15]	KNN	Breast Cancer Data in Gaza Strip BCDG	0.76	N/A	N/A
	SVM		0.77	N/A	N/A
	ANN		0.75	N/A	N/A
Present work	k-NN	Breast Cancer Wisconsin dataset	0.99	1.00	0.98
	SVM		0.94	0.98	0.80
	DT		0.89	0.92	0.88
	ADA		0.93	1.00	0.92

V. CONCLUSION AND PERSPECTIVES

This paper presents an overview of some data mining techniques; ANN, k-NN, Binary SVM, and AdaBoosting. The four CAD techniques applied to the Breast Cancer Wisconsin Dataset; groups together 32 statistical attributes between inputs and the output. The data analysis and classification are based on the confusion matrix to extract the number of FN, TP, TN, and FP. The results show that k-NN is more reliable

to assist in decision making regarding breast cancer severity with this kind of dataset after features selection.

CAD systems are always open to further research to develop and optimize the algorithms to improve the system and making it more accurate, more specific, and way more sensitive.

Studies tend to migrate to Convolutional Neural Network CNN application on breast cancer images. Further work along

this line should focus on combining classifiers to improve their performance to reach the goal of a more precise and accurate classification.

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